Keeping the Referent in the DARK: Anchor-Preserving Token Compression for Privacy-Aware Inference on IoT Logs

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Abstract

IoT device logs are long, identifier-dense, and routinely copied into language-model prompts. A question about a device or an account names exactly the value that must not reach the model: the queried entity is also the entity under protection. Redaction removes the value and, with it, the handle the model needs to answer. Generic compression drops context and frequently drops the record that carries the answer. We present DARK, a training-free routing rule that decides every context unit with one of four operations—delete, alias, refill, keep—and treats the sensitive-and-relevant cell differently from the rest: the unit stays in the prompt, its protected values are replaced by stable aliases, and the same alias map is applied to the question. Relevance is read off the question alone—the record block that mentions the queried entity—so the compressor never consults the reference answer. On a 112-case benchmark over six public datasets, DARK reaches F1 0.6366 using 48.6% of the original tokens with no protected value recoverable from the prompt, and is statistically indistinguishable from aliasing the full context without compression (0.6475; difference -0.0109, 95% CI $[-0.0824, +0.0612]$). On a 2,728-case extension it matches the strongest query-aware compressor (0.4937 vs. 0.4831; difference not significant) while leaving 0% of protected values recoverable against that method's 98.1%, and it exceeds the only leak-free baseline by 0.3226 F1 points. The same ordering holds at every budget from 10% to 75%, on a second model, and under two question paraphrases.

1 Introduction

Operational logs from consumer IoT deployments record what each device did, when, and for which account. They are long (an excerpt in our benchmark reaches $19{,}838$ tokens), repetitive, and full of identifiers: account handles, device serials, contact fields. They are also a natural substrate for question answering. An operator asks which device is associated with a given account and expects a short answer.

The property that makes such a question answerable is also what makes it unsafe to send verbatim. The question names an entity, and that entity is itself a protected value. Redacting it before inference is the standard remedy, and it destroys the task: once the account handle is gone, the model cannot tell which of two hundred similar records the question refers to. Compressing the context is the other standard remedy, and it destroys the task for a different reason. Under a redundancy-ordered rule the record carrying the answer survives compression in 47.5% of cases on our benchmark, and in only 37.2% of needle cases, where the answer value repeats across distractor records and the score gives it no priority. Both remedies treat protecting the value and keeping the reference as competing goals to be balanced.

For this class of query, they need not compete. Figure 1 states the conflict and the two ways of failing it; the rest of this paper is about the third option. The observation that makes this possible is layout. Records in these excerpts are grouped into blocks that hold an entity's identifying line together with the attributes recorded for it, so the queried entity and the answer to the question about it sit in the *same*

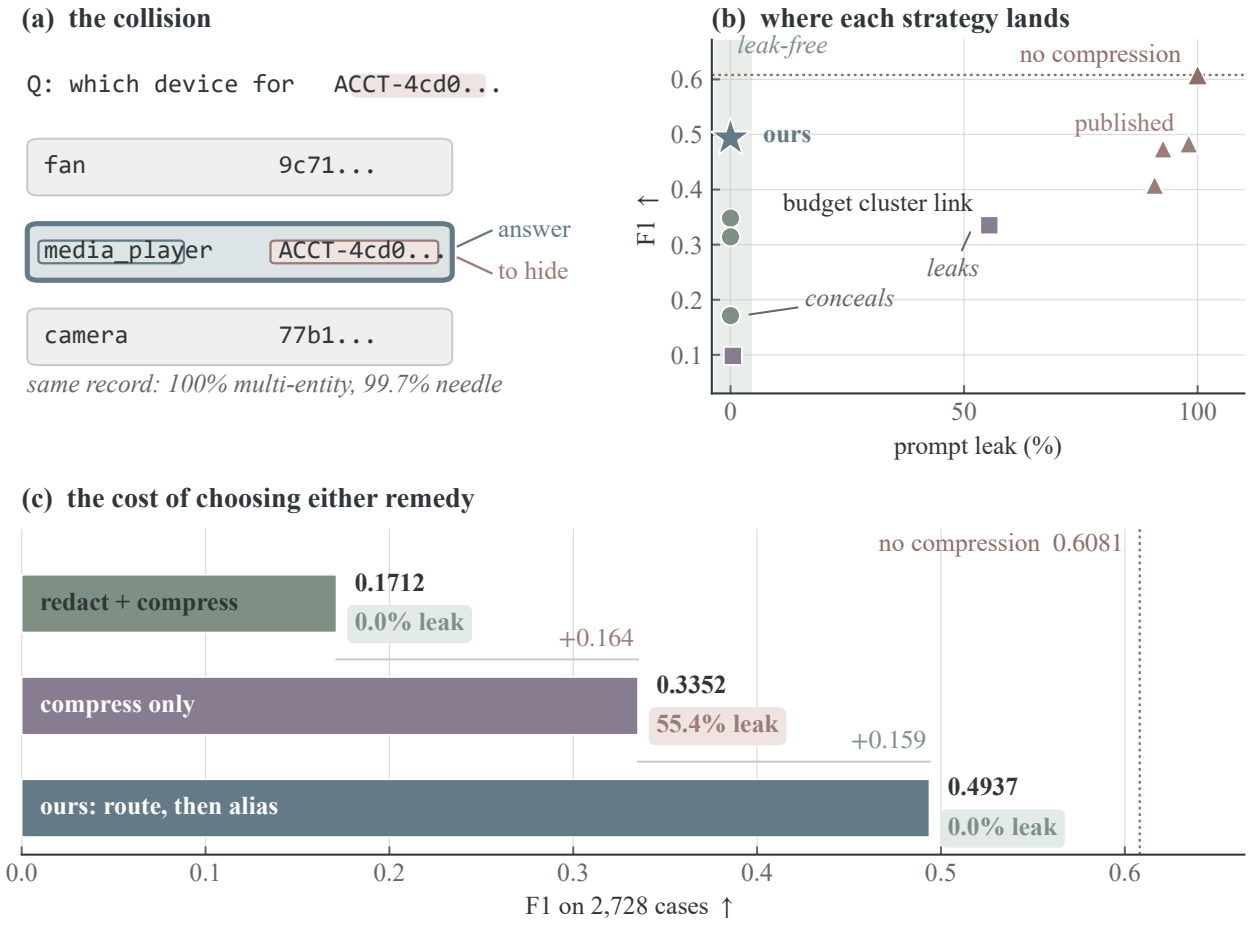


Figure 1: The conflict, and where each strategy lands on it. (a) A question names an account, and the record that answers it also stores that account as a value to hide: the referent and the protected span coincide. (b) F1 against prompt leakage for ten arms on the 2,728-case extension. Two clusters fail in opposite directions—the leak-free arms that drop protected spans lose the referent (0.098 to 0.348), and the published compressors forward the value in 90.8% to 100% of cases. Only the marked point reaches 0.494 without leaking. (c) The two standard remedies against the routing rule; labels give the F1 gained at each step, and the pills give the leakage that comes with it.

block in 100% of multi-entity cases and 99.7% of needle cases. A pipeline that preserves that block, and replaces the protected values inside it with stable aliases, keeps a referential handle while never exposing the value. Finding the block requires no answer key: the question already names the entity, and the block is whatever contains that mention.

What this implies for the design. A single global decision cannot express it. Aliasing every protected value keeps the handle but spends budget on sensitive records that no question component will use; deleting every protected value removes the handle along with the value. The two axes have to be decided per unit, and the sensitive-and-relevant cell needs an operation that neither redaction nor deletion provides. We therefore route each unit on the pair (sensitivity, relevance) and give each cell its own opera-

tion. Sensitive and irrelevant units are *deleted*. Irrelevant insensitive units *refill* whatever budget remains. Relevant insensitive units are *kept* verbatim. Sensitive-and-relevant units are *aliased* rather than deleted. The four operations give the method its name, DARK: **D**elete, **A**lias, **R**efill, **K**eep.

Evidence preview. On a 112-case benchmark spanning six public datasets, DARK uses 48.6% of the original tokens, leaves no protected value recoverable from the prompt, and scores F1 0.6366. Aliasing the whole context without compressing it scores 0.6475 on the same cases, a difference that is not significant; not compressing at all scores 0.7189. The gap that matters is against pipelines that keep the same privacy guarantee. The strongest of those reaches 0.4308, and in the multi-entity subset it still exposes protected values. On a $2{,}728$-case extension, DARK is level with the best query-aware compressor (0.4937 vs. 0.4831) at 0% leakage against 98.1%. Removing the alias-instead-of-delete rule costs 0.1795 F1 points overall and 0.3273 on multi-entity cases. On the topical-question variant, where no entity can be located, it costs $+0.0013$ and is not significant. The rule acts where it is designed to act and is inert elsewhere.

Contributions.

- **A failure mode that the compression literature does not report.** When the queried entity is also the protected object, redundancy ordering carries almost no signal about which record the question is about. We quantify it on $2{,}728$ cases (47.5% retention overall, 37.2% on needle cases, both at a budget cut that would give an uninformative rule 50%) and show that protecting the anchor *line* alone repairs only half of the gap (57.6%).
- **A training-free routing rule that does.** Sensitive-and-relevant units are aliased rather than deleted, and relevance comes from the question by locating the record block that contains the queried entity. The rule needs no reference answer, no training, and no model access.
- **An evaluation that separates privacy from utility at equal budget.** Across two benchmark sizes, three public-benchmark variants, five budgets, two models, and two paraphrases, DARK is compared with twelve published compressors and sanitizers re-run on the same cases, with paired bootstrap intervals on every difference we report.

Section 2 positions the work; Section 3 fixes notation and the threat model; Section 4 presents the routing rule; Sections 5–7 describe the benchmark and the results; Section 8 isolates each design decision; Section 9 states the scope the evidence supports.

2 Related Work

Prompt compression. One family reduces prompt length by scoring and removing tokens or sentences. LLMLINGUA [6] and LONGLLMLINGUA [7] use a small language model to estimate token informativeness; LLMLINGUA-2 [10] distils the decision into a token classifier for task-agnostic compression; Selective Context [8] drops spans that carry little self-information; Provence [2] prunes sentences jointly with question relevance; CPC [9] scores sentences with a context-aware encoder; Quito-X [16] casts compression as an information-bottleneck problem and approximates mutual information with cross-attention; DAC [18] adds an attention-aware term to an entropy criterion; the Perception Compressor [15] retrieves relevant demonstrations with guiding questions before compressing at the token level. These methods differ mainly in how they estimate relevance and redundancy. They share one assumption: that the prompt is ordinary text whose only cost is length. None of them alters a value the model is allowed to read, and none is evaluated on questions whose answer *is* a protected value. The omission is consequential. The strongest of these methods leaves 98.1% of protected values recoverable in our benchmark, and it preserves the answer for a query that names a sensitive entity far less reliably than it preserves an ordinary answer span.

Text sanitization for inference. A second family rewrites protected spans before the prompt leaves the user side. DYNTEXT [17] substitutes sensitive tokens with semantically similar non-sensitive neighbours drawn from a dynamic adjacency list, so that a fixed substitution table

cannot be attacked; PromptGraph [4] guides the rewrite with a dependency graph; INTACT [11] replaces identifiers with generalised candidates selected for resistance to LLM inference attacks. Masking wrappers around published compressors belong here too. These methods decide per span what to hide and treat utility as whatever survives the rewrite. On our benchmark that framing has a measurable cost: a masking pipeline that achieves zero leakage scores 0.1712 on the $2{,}728$-case extension, because the span it removes is the referent the question needs. Differential privacy [3] and k-anonymity [14] supply formal guarantees, but neither offers the per-query referential behaviour this task requires; Section 9 returns to that point.

Long-context question answering. LongBench [1] supplies the passages from which we derive the public-benchmark variants, and SQuAD v2 [13] supplies the scoring function. Our contribution is orthogonal to the retrieval and long-context modelling literature: we do not modify the reader, only the text it receives, and we measure the effect at a fixed token budget.

Where this study sits. The two families above optimise different objectives, length and concealment, and each reports utility on prompts whose answer is ordinary text. We study the case where both objectives fall on the same span, and show that the resulting conflict is resolvable by changing *which operation* acts on that span rather than by trading one objective against the other. We keep the compression interface of the first family and the per-span rewriting of the second, and modify the stage at which the operation is chosen.

3 Problem Formulation

Setting. A prompt is a pair (q, C): a question q written on the user side and an excerpt C drawn from an IoT log. The excerpt is split into units $U(C) = \{u_1, \ldots, u_m\}$, one per record or key–value group. Each case carries a set of protected values $V = \{v_1, \ldots, v_n\}$; a unit is *sensitive* when it contains a value in V. A model M answers q from the compressed context, and the answer is scored against a reference span.

What must not reach the model. The prompt is the object of study. It travels from the device to the model process and may be logged, cached, or inspected. A policy is *leak-free* on a case when no value in V is recoverable from the exact text handed to the model, that is, from the rewritten question q' concatenated with the compressed context, not from the context alone. Recoverability is measured with the pattern set of the published protocol (full value plus fragments). We assume no adversarial manipulation of the model and no side channel beyond the prompt text, and we do not model an adversary who holds the alias map; Section 9 states what that leaves out.

The queried entity. Each question names an entity $e^\star$, which may be an account, a device, or a record identifier. The *anchor* $A \subseteq U(C)$ is the set of units containing a mention of $e^\star$. A unit plays a referential role when the question cannot be resolved without knowing which entity a value belongs to.

Budget. A budget arm receives a ratio r and must realise a compressed context of at most $B = \lceil r \cdot n_C \rceil$ tokens, where n_C is the token length of the uncompressed excerpt under the tokenizer used to build the prompt. Exceeding B is a hard error rather than a silent truncation, because in this domain the answer region is often at the end of the excerpt and truncation would remove it without notice.

Metrics.

- **Task quality.** EM and F1 from the SQuAD v2 scorer [13]; UNKNOWN is not a legal answer.
- **Prompt leak**. Fraction of cases in which any protected value is recoverable from question plus compressed context. Lower is better.
- **Gold-in-context (GIC)**. Fraction of cases in which the reference span survives compression. It is model-free, and therefore an upper bound on what the reader can achieve; it is also blind to whether the surviving span is still interpretable, which is why we report task quality as a separate endpoint.
- **Overshoot**. Indicator that a budget arm exceeded B on a case.

The condition we study. The setting is narrow and we state it as such: long, identifier-dense operational text; a question that names an entity; and a requirement that the entity's value not reach the model. Questions of this form are common in IoT operations and, as Section 8 shows, are exactly the ones on which redundancy-based compression is harmful. The same routing reduces to ordinary compression when the question is topical and names no entity; we report that variant as out of scope rather than as a success.

4 Method

4.1 Overview

Given an excerpt C, a question q, a budget ratio r, and the set V of protected values, DARK produces a rewritten question q' and a compressed context P such that (q', P) fits the token budget and contains no value of V. The pipeline has four steps: split C into units; label each unit on sensitivity and on relevance; apply the operation attached to its cell; and fit the surviving units into the budget, applying the alias map to the question at the end. The four operations give the method its name: **D**elete, **A**lias, **R**efill and **K**eep. Algorithm 1 states the procedure. There is no training and no model access at this stage. The only deployment-specific input is V, which an operator already holds for redaction. Figure 2 shows the pipeline end to end and the evidence for its one non-obvious choice.

4.2 Relevance from the question alone

The relevance label decides which region must survive and which may be spent on the budget. We use a single signal: the region containing a mention of the entity the question names. The signal takes two forms in our excerpts.

Record blocks. When the excerpt is segmented into blocks headed by a record marker, the whole block that contains the anchor mention is relevant. This is the common case for log-shaped benchmarks: 100% of multi-entity cases and 99.7% of needle cases place the reference answer inside the same block as the anchor mention. A rule defined on blocks therefore keeps the answer without inspecting it.

Algorithm 1 DARK prompt construction

Require: excerpt C, question q, budget ratio r, protected values V

$U \leftarrow \textsc{Split}(C)$
$A \leftarrow \{u \in U : u \text{ mentions } e^\star(q)\}$
$A \leftarrow \textsc{RecordBlock}(U, A)$ ▷ if C has record markers
for all $u \in U$ **do**
  if $u \in A$ **then**
    if u is sensitive **then**
      alias(u) ▷ sensitive and relevant
    else
      keep(u) ▷ relevant only
    end if
  else if u is sensitive **then**
    delete(u) ▷ sensitive and irrelevant
  else
    mark u as refill
  end if
end for
$P \leftarrow \textsc{Fit}(\text{kept} \cup \text{aliased}, \text{refill}, B)$ ▷ hard B; degrade rather than fail
$q' \leftarrow \textsc{AliasQuestion}(q)$ with the same map
return (q', P)

Sentence windows. LongBench-style passages carry no block markers. There the relevant region is the anchor sentence plus one sentence on each side: five units, about 2.3% of the budget in our measurements. The window is a fallback, not the main mechanism, and Section 8 reports where it suffices and where it does not.

No answer key. A natural implementation of a relevance label is an oracle: mark the unit that contains the reference answer and deprioritise it last when dropping units. That implementation is not deployable, and it changes what the method is. We avoid it, and Section 8 measures how much a pipeline gains from having it, so that the comparison against an oracle-assisted pipeline is explicit rather than implicit.

4.3 Four cells, four operations

Sensitivity is a property of the unit given V: a unit is sensitive when it contains a protected value. Crossing

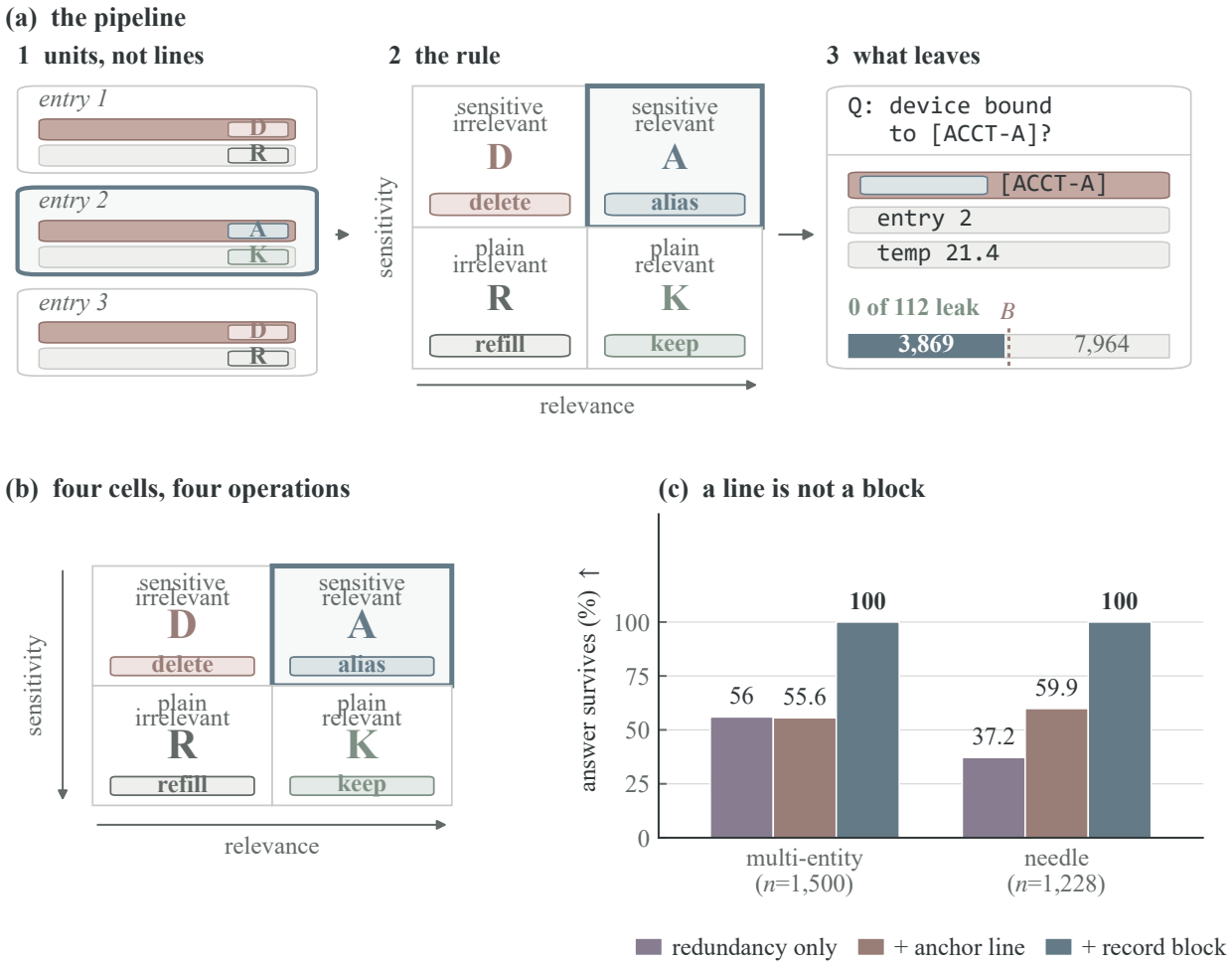


Figure 2: The pipeline. (a) An excerpt is split into units inside record blocks; the block that mentions the entity named by the question is the relevant region, and every unit is then routed on (sensitivity, relevance) before the surviving set is fitted to the budget. The strip at the right reports a real 112-case run: 3,869 tokens against 7,964, and no protected value in any prompt. (b) The four cells and the operation attached to each. The top-right cell is the one that keeps a protected value addressable instead of removing it; the numbers on its right give the measured cost of replacing that operation by deletion. (c) Protecting the anchor *line* does not close the gap; protecting the *block* does. Bars give the fraction of the 2,728 cases in which the answer-bearing record survives.

the two labels gives four cells, and each cell receives the operation that lets the reader resolve the question while the value stays out of the prompt.

Sensitive and relevant: alias (A). The unit remains in the context, and each of its protected values is replaced by a stable alias from a per-run mapping, so that repeated mentions of one entity receive one token. This cell is where we depart from prior practice. Redaction, the standard primitive, gives the same concealment but removes the referent. Keeping the unit and renaming its values preserves the association between the entity and its attributes while satisfying the constraint on V. Section 8 reports the cost of replacing this operation by deletion: 0.1795 F1 points

overall and 0.3273 on multi-entity cases.

Sensitive and irrelevant: delete (D). The unit holds a protected value that no question component refers to. It contributes nothing to the answer, so it is removed rather than aliased. This is the only cell in which a protected value disappears from the prompt entirely.

Relevant and insensitive: keep (K). The unit bears on the question and holds nothing that must be hidden, so it is kept verbatim.

Irrelevant and insensitive: refill (R). Remaining units hold no protected value and lie outside the relevant region. They compete for whatever budget is left, ordered by the redundancy score of Section 4.5.

4.4 Question-side aliasing

Aliasing the context alone produces a prompt whose two halves use different names for the same entity: the context refers to `[ACCOUNT-A]` while the question still names `ACCT-4cd02ed8fb3a`. The reader cannot connect the two, and a string-presence metric reports the entity as preserved while the task is destroyed. We therefore rewrite the question with the same map, so that every mention of the queried entity carries the alias used for it inside the context. The object handed to the model is the pair (q', P), and both halves are written with one map. Section 8 measures the contribution of this step.

The alias assignment also determines whether a reader can invert the map by rank. We fix it per run from a keyed permutation instead of by value order; an unkeyed assignment lets an adversary who knows the value set recover every alias from its rank alone. All reported numbers use the keyed assignment.

4.5 Budget accounting

The budget is $B = \lceil r \cdot n_C \rceil$, and it is enforced rather than approximated. The surviving set is assembled greedily and then trimmed against the *actual* concatenated text, because the separator between units costs tokens: a sum of per-unit token counts is a lower bound on the length of the joined prompt. When the trimmed set still exceeds B, units are discarded in a fixed priority—refill first, then non-anchor relevant units, then anchor units—so that the pipeline degrades instead of failing. Every budget arm reported in this paper has zero overshoot.

Refill ordering uses a model-free redundancy score: for unit u_i, the maximum Jaccard similarity between its token set and that of any other unit. Candidates are visited in *increasing* redundancy, which keeps the repetitive records.

That direction deserves comment, because it inverts the usual reading of the score. Visiting units in decreasing redundancy and excluding them is the principled interpretation of a redundancy measure, and it is what a standard budget-fitting routine does. Here it is the wrong choice: the answer-bearing record is frequently the most redundant unit, because the answer value repeats across distractors. Section 8 reports both variants and the 0.6567-point difference they produce on the two-hop benchmark. We state the choice as an empirical regularity with a stated mechanism, not as a consequence of the definition.

5 Benchmark

Construction. We build the benchmark from six public IoT and smart-home corpora that appear in the retrieval and long-context literature. For each source we assemble excerpts in which records for several entities of the same type are interleaved, and we inject synthetic protected values (account identifiers, contact addresses, phone numbers) into the records that already exist, rather than into separate carriers. Reference answers are derived deterministically from the injected structure, so no answer is authored by hand.

The injected values are canaries, not real personal data: public corpora cannot carry real PII, and using a canary makes leakage measurable exactly. The consequence is that our privacy claim concerns the recoverability of *these* values, not of arbitrary personal information that may co-occur in real logs.

Question form. Each case poses an entity-resolving question: the question names an account and asks for an attribute recorded for it (which device is associated, which persona identifier is stored, and so on). Two families of excerpts are included. In *multi-entity* excerpts the same question template applies to six accounts and the target

account is distinguished only by its identifier. In *needle* excerpts the question names a specific record identifier and asks for a field value inside it.

Sizes. We report three sets. The *main* set contains 112 cases, 20 from each of five sources and 12 from a sixth, and is used for the main table, the budget curve, the cross-model and the paraphrasing experiments. The *extension* set contains 2,728 cases from the same six sources and is used to check that the main-set conclusions survive an order-of-magnitude increase in size and a different source mix. Both sets share the same question templates.

A note on key hygiene: the two sets reuse the same case identifiers for different content, and joining results across them by identifier silently mixes records. Every analysis reported here (i) resolves results within a single set, and (ii) verifies the join by comparing the uncompressed-context arm against a reference value computed from the same set. The check is reported alongside each table.

Public-benchmark variants. To test the rule outside our own construction we derive three variants from LongBench passages, 300 cases each, with the same injected canaries. *Topical* leaves the original questions, which name no entity. *Single-hop* replaces the question with one that names an account, and places the answer in the same record as the account. *Two-hop* places the answer six sentences away from the account mention, so that resolving the question requires following the record structure rather than the local window. Section 8 reports all three, including the one we do not claim.

6 Experimental Setup

Models and decoding. The main reader is Qwen2.5-7B-Instruct [12]; the cross-model check uses Mistral-7B-Instruct-v0.3 [5]. Both run in fp16 with greedy decoding, 64 new tokens, maximum context 24,576 tokens and a strict policy that aborts rather than truncates when a prompt exceeds it. The random seed is fixed at `20261015`. Scoring uses the official SQuAD v2 script [13].

Baselines. We re-run eight published compressors through their reference implementations: LLMLingua-2 [10], LongLLMLingua [7], Selective Context [8], Provence [2], CPC [9], Quito-X [16], DAC [18], and the Perception Compressor [15]. Where a method exposes both a budget-matched and a native configuration we report both; the budget-matched one is used for the headline comparison, since published defaults vary in length. We additionally run two sanitization-oriented baselines, INTACT [11] and a masking wrapper around LLMLingua-2, and our own non-routing arms: `ful`, `alias` (alias all protected values, no compression), `delete` (redact all, no compression), `alias_budget` and `alias_budget_link` (global alias, then redundancy-based selection without and with anchor protection), and the corresponding `budget_cluster` pair on raw text.

What each baseline receives. All compressors receive the question and the context and nothing else. We verified in the runner that the reference answer is never passed to any baseline, and that no baseline consumes the anchor set. This is the fairness condition for the comparison: the contribution we claim is precisely the ability to obtain the referential benefit without the answer key.

Equal-budget comparison. Budget arms are anchored at $r = 0.5$ for the main comparison and swept over $r \in \{0.10, 0.15, 0.25, 0.50, 0.75\}$ for the budget curve. Published compressors are evaluated at their own 50% setting, with the token accounting reported in every table so that a method that realises a different length is visible as such.

Statistics. All comparisons between two arms on the same cases are paired. We report the mean paired difference and a 95% percentile bootstrap interval with $10{,}000$ resamples over cases. A difference is called significant when the interval excludes zero. We deliberately do not attach a significance claim to any comparison across the two benchmark sizes, which draw different cases.

Reproducibility controls. Two controls are run in every batch. First, the uncompressed arm is recomputed; it involves no selection logic, so it must reproduce the reference value, and any deviation indicates that the run differs in model, precision, or data. Second, every budget arm

Table 1: Main comparison on the 112-case benchmark (Qwen2.5-7B-Instruct, $r = 0.50$, seed 20261015). All arms of ours are constructed without access to the reference answer. *tokens* is the mean compressed-context length; *leak* counts cases in which a protected value is recoverable from question plus context. Arrows give the preferred direction. BCL = `budget_cluster_link`, ABL = `alias_budget_link`, ALB = `alias_budget`, BC = `budget_cluster`. Controlled comparison is the first three rows: compressing to 48.6% of the original length costs 0.0109 against the no-compression aliasing arm, which is not significant. The uncompressed arm is the reference value used to verify that a result set belongs to this benchmark.

Arm	F1 ↑	tokens ↓	leak ↓
`full`	0.7189	7964	112/112
`alias`	0.6475	7824	0/112
DARK	**0.6366**	**3869**	**0/112**
`delete`	0.4916	7845	0/112
BCL	0.4308	3683	47/112
ABL	0.4219	3638	0/112
ALB	0.2232	—	0/112
BC	0.2143	—	8/112

Paired differences against DARK: `alias` -0.0109 $[-0.0824, +0.0612]$ (n.s.); `full` -0.0823 $[-0.1610, -0.0041]$; ABL $+0.2147$ $[+0.1224, +0.3085]$.

asserts non-overshoot at the case level. Runs that fail either control are discarded rather than reported.

7 Results

7.1 Compression costs little

Table 1 answers the first question: at an equal 50% budget, does the routing rule preserve the task? Aliasing the whole context without compressing scores 0.6475; DARK scores 0.6366 at 48.6% of the tokens. The paired difference is -0.0109 with a 95% interval of $[-0.0824, +0.0612]$, which does not exclude zero: on this benchmark, halving the context costs no measurable task quality relative to the leak-free uncompressed arm. Against not compressing and not sanitizing, the deficit is -0.0823 $[-0.1610, -0.0041]$, which is the price of the privacy constraint rather than of compression, since the aliasing arm pays it as well (-0.0714 against `full`).

The comparison specific to our design is the lower block of the table. Two arms are leak-free and compress to nearly the same length: `alias_budget_link`, which aliases globally and protects the anchor line, and DARK. The difference is $+0.2147$ $[+0.1224, +0.3085]$. The two do not differ in the privacy operation—both alias—but in which region survives: one line, or the block containing it.

7.2 At scale, the comparison is a privacy comparison

Table 2 reports the $2{,}728$-case extension. DARK scores 0.4937, level with the strongest query-aware compressor (Quito-X, 0.4831; difference $+0.0107$, interval $[-0.0091, +0.0306]$) and above CPC (0.4740) and Provence (0.4080). What separates the arms is leakage. Every published compressor leaves protected values in the prompt, in 88.8% to 100% of cases, because concealing them is not among its objectives. DARK leaves none. Among leak-free arms the nearest competitor is a masking wrapper at 0.1712, which DARK exceeds by $+0.3226$ $[+0.3041, +0.3414]$.

The reading we take from this table is narrow. A query-aware compressor reaches the same F1 as our rule at the same budget, but only by forwarding the value the question names. When that value must not be forwarded, the compressor loses the task, as the masking row shows, and the routing rule does not.

7.3 The advantage holds across budgets

Table 3 varies the budget over five ratios. DARK is the strongest budget arm at every point: $+0.205$ at $r = 0.10$, $+0.263$ at $r = 0.25$, $+0.215$ at $r = 0.50$, and $+0.065$ at $r = 0.75$, all at zero leakage. Its own score is not monotone. It peaks at $r = 0.50$ and drops by 0.021 at $r = 0.75$, because the additional refill text admitted at a loose budget is repetitive context that competes with the relevant block for the reader's attention. The competing arms keep improving with budget, so the margin narrows; at $r = 0.75$ it is small. We report the non-monotonicity rather than describing the curve as monotone, and note that the half-budget setting used elsewhere is where the mechanism separates most clearly from the alternatives.

Table 2: Comparison on the 2,728-case extension (Qwen2.5-7B-Instruct, $r = 0.50$). Published compressors are re-run through their reference implementations on the same cases; arms suffixed `_native` use each method's own default length. *Leak* is the fraction of cases in which a protected value is recoverable from the prompt. DARK and the masking baseline are the only arms at 0%; DARK exceeds that baseline by $+0.3226$ $[+0.3041, +0.3414]$. Against the strongest compressor, Quito-X, the difference is $+0.0107$ $[-0.0091, +0.0306]$, i.e. not significant, at 0% versus 98.1% leakage. The uncompressed arm reproduces the reference value 0.6078 from a separately evaluated batch to within one case in $2{,}728$.

Method	Family	F1 ↑	leak ↓
`full`	uncompressed	0.6081	100.0%
`ext_full`	uncompressed (rerun)	0.6078	100.0%
DARK	ours (routes)	**0.4937**	**0.0%**
Quito-X	query-aware	0.4831	98.1%
Quito-X `_native`	query-aware	0.4862	98.1%
CPC	query-aware	0.4740	92.6%
CPC `_native`	query-aware	0.4523	93.4%
Provence	query-aware	0.4080	90.8%
Provence `_native`	query-aware	0.4047	88.8%
`alias_budget_link`	ours, no answer key, merged cell	0.3482	0.0%
`budget_cluster_link`	raw text, anchor kept	0.3352	55.4%
`quad4_del`	ours, delete instead of alias	0.3142	0.0%
LLMLingua-2	task-agnostic	0.2651	100.0%
LLMLingua-2 `_native`	task-agnostic	0.2433	100.0%
`mask_llmlingua2`	sanitize + compress	0.1712	0.0%
`mask_llmlingua2_native`	sanitize + compress	0.1663	0.0%
`alias_budget`	global alias, no anchor	0.1469	0.0%
`budget_cluster`	redundancy only	0.0981	0.5%

Table 3: Budget curve on the 112-case benchmark. Every row satisfies its budget with zero overshoot. DARK is the strongest budget arm at all five ratios; its advantage narrows as the budget grows, and its own score peaks at $r = 0.50$ because at $r = 0.75$ the additional refill text is noise. The last row gives the mean realised context length in tokens.

Arm	@0.10	@0.15	@0.25	@0.50	@0.75
DARK	**0.4357**	**0.4681**	**0.5116**	**0.6366**	**0.6158**
`quad4_del`	0.3297	0.3788	0.3955	0.5205	0.5086
BCL	0.2304	0.2052	0.2310	0.4308	0.5576
ABL	0.2304	0.2052	0.2488	0.4219	0.5512
realised tokens	656	1033	1913	3869	5830

Uncompressed reference: F1 0.7189 at 7964 tokens.

7.4 Cross-model and phrasing stability

On Mistral-7B-Instruct-v0.3 at $r = 0.50$, DARK scores 0.6946 against 0.6366 on Qwen2.5-7B. Every other arm loses ground when the reader changes: `full` -0.0258, `alias` -0.0544, `delete` -0.0357, `alias_budget_link` -0.0471, `budget_cluster_link` -0.0471. The routing rule is the only arm that improves, which is consistent with the mechanism: a second reader gains more from a cleaned, budgeted prompt than from a long one.

Against two mechanical rewritings of the question, DARK scores 0.6637 and 0.6023 against 0.6366 on the original, a maximum change of -0.0343. The uncompressed arm changes by -0.0614 over the same rewritings. The margin over `alias_budget_link` remains significant in all three conditions ($+0.2147$, $+0.2711$, $+0.1994$). The check is relevant because the rule keys on a mention of the queried entity, and a rewriting could in principle disturb that mention; in these variants it does not.

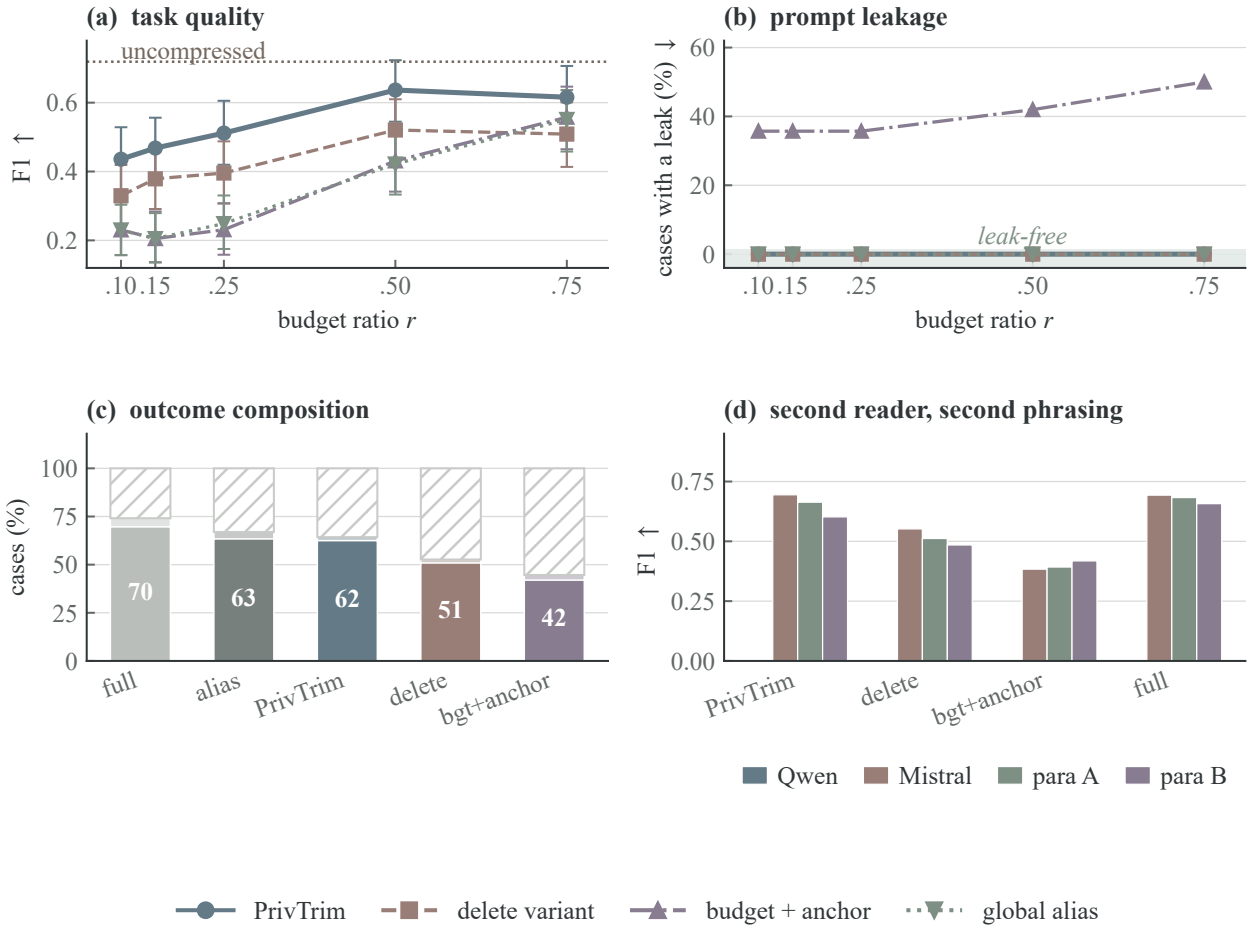


Figure 3: Behaviour across the budget sweep (112 cases). (a) F1 at five ratios with bootstrap 95% intervals over cases; the dotted line is the uncompressed reference. (b) Fraction of cases in which a protected value is recoverable from the prompt. DARK, the delete variant and the global-alias arm stay at zero for the whole range, while the arm that compresses raw text leaks in 35.7% of cases at tight budgets and 50.0% at $r = 0.75$, so its apparent parity at loose budgets is bought with exposure. (c) Composition of case outcomes at $r = 0.50$: solid is an exact answer, the pale band is a partial match, hatched is a wrong or UNKNOWN answer. (d) The same four arms under a second reader and two question paraphrases; the ordering of the arms is preserved in all four conditions.

8 Analysis

8.1 Why redundancy ordering fails here

The failure we set out to address is not visible in aggregate F1, so we state it in its own terms. The question is whether the record carrying the answer survives compression when the rule that decides is a redundancy ordering. On the 2,728-case extension the redundancy-only rule retains the answer record in 47.5% of cases (56.0% multi-entity, 37.2% needle), protecting the anchor line raises this to 57.6%, and routing on the record block reaches 100%.

The two failure rates say different things. Multi-entity survival sits either side of the budget cut, which is what an uninformative ordering produces: the score does not know which record the question is about, so it keeps the right one about half the time. Needle survival is *below* the cut, and

Table 4: Isolating the alias-instead-of-delete rule. Replacing the alias operation by deletion in the sensitive-and-relevant cell, with everything else held fixed, costs a large and significant amount wherever the anchor is itself a sensitive value (multi-entity, single-hop, two-hop) and nothing where it is not (topical variants, no anchor). The two-hop column is the clearest case: the reference answer is present in the context in 96.3% of cases under the delete variant, and F1 is nevertheless 0.0000, because the account that identifies the record has been removed.

Benchmark	DARK	delete variant	Δ	95% CI
extension, all (2,728)	0.4937	0.3142	$+0.1795$	$[+0.1643, +0.1945]$
multi-entity (1,500)	—	—	$+0.3273$	$[+0.3040, +0.3513]$
needle (1,228)	—	—	-0.0010	$[-0.0105, +0.0086]$ (n.s.)
main set (112)	0.6366	0.5205	$+0.1161$	$[+0.0446, +0.1875]$
LongBench topical (300)	0.3516	0.3503	$+0.0013$	$[-0.0002, +0.0042]$ (n.s.)
LongBench single-hop (300)	0.9607	0.1674	$+0.7933$	$[+0.7430, +0.8426]$
LongBench two-hop (300)	0.6733	0.0000	$+0.6733$	$[+0.6200, +0.7256]$

there the ordering is worse than uninformative. The answer in a needle case is a field value that recurs across distractor records, so the record holding it looks like a typical record and inherits no priority from the score. Protecting the anchor line repairs this for needle cases—retention rises from 37.2% to 59.9%—but the line it protects is not the line that carries the answer, which is why the repair stops short of 100% rather than reaching it.

Figure 4 breaks the same question down by budget, by source, and by what the retention is worth. Panel (a) follows the answer record across the budget curve: the two arms that route on the record block hold it in 100% of the 112 cases from $r = 0.50$ upward and in 56.2% at $r = 0.10$, while the two arms that only protect the anchor line stay between 24.1% and 77.7% across the whole range. Panel (b) sets survival against F1 for the 2,728-case extension. Survival accounts for a good part of the ordering but not all of it: the delete variant keeps the answer record in 100% of cases and still scores 0.0000 on multi-entity cases, because the record it keeps no longer says which account it belongs to. The dissociation is not an artefact of the aggregate: panel (c) repeats the comparison within each of the six sources, and the block-level rule holds the answer in every case of every source while the redundancy-only rule falls to 16.7% on `toolmem_raw`. Panel (d) shows what that retention is worth per source; the one source where the block rule is behind, by -0.065, is the one where the redundancy-only arm retains 83.3% and the two are effectively tied.

8.2 Protecting the anchor line is not enough

A natural repair forbids dropping units that mention the queried entity. It does not close the gap, and where it helps it helps for the wrong reason. Overall retention moves from 47.5% to 57.6%, but the two families behave differently: multi-entity retention is unchanged (56.0% to 55.6%), while needle retention rises from 37.2% to 59.9%. Needle cases improve because the anchor there is the record identifier, which happens to share the record with the answer far more often than the account line does. The reason the repair stops short of 100% is layout: the line carrying the account identifier and the line carrying the answer are different units in 100% of multi-entity cases and in 65.4% of needle cases. Preserving the identifier preserves the pointer, not the content.

The repair becomes effective one level up. The excerpt is laid out in record blocks, and the block containing the anchor mention also contains the answer in 100% of multi-entity cases and 99.7% of needle cases. Constraining the block instead of the line restores answer retention to 100% on both families without consulting the answer key. This is the measurement that motivates the routing rule: the relevant region is a block, not a unit.

8.3 How much a pipeline gains from the answer key

Several published pipelines are evaluated with a rule that receives the answer span. Since we claim to obtain the referential benefit without it, the size of that benefit belongs

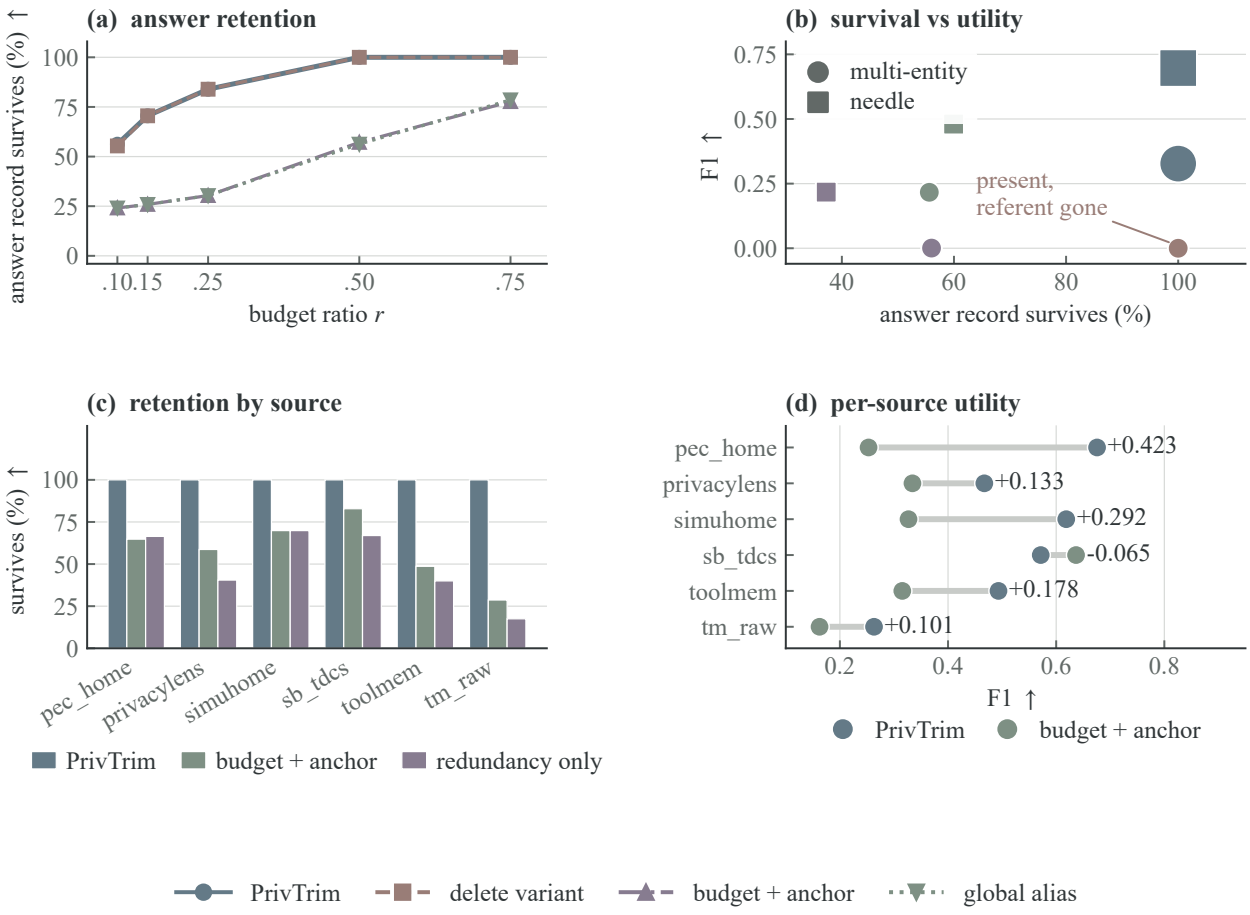


Figure 4: Answer retention and what it does not explain. (a) Fraction of the 112 benchmark cases in which the record carrying the reference answer survives compression, as the budget varies. DARK and the delete variant coincide almost exactly: the alias-instead-of-delete rule does not change *which* units survive. (b) Survival against F1 on the 2,728-case extension; circles are multi-entity cases and squares are needle cases. The delete variant sits at 100% survival and 0.0000 F1 on multi-entity cases: the information is in the prompt and cannot be located. (c) Retention within each source at $r = 0.50$. (d) Per-source F1, DARK against the anchor-protected arm, with the paired difference printed at each row.

on the record. On a 300-case stratified sample, enabling the answer key changes F1 as follows.

Arm	Δ F1 from answer key ↑	95% CI
BC	+0.2617	[+0.215, +0.310]
ALB	+0.2850	[+0.233, +0.337]
BCL	+0.5583	[+0.505, +0.613]
ABL	+0.5917	[+0.535, +0.648]

The gain is largest for the anchor-protected arms, which are the arms used in the main comparison. Reporting such

a number without the corresponding no-key value would overstate what a deployed pipeline can do. The tables in Section 7 report the no-key values; this entry is here so that the two can be compared.

8.4 Alias, not delete

Table 4 and Figure 5 hold everything fixed except the operation applied in the sensitive-and-relevant cell, and the pattern is the one the design predicts. Where the anchor is itself a sensitive value, the difference is large and significant: $+0.3273$ on multi-entity cases, $+0.7933$ on the single-hop variant, $+0.6733$ on the two-hop variant. Where no anchor exists—the topical variant, whose questions name no entity—it is $+0.0013$ and not significant. On the needle family, whose anchor is a non-sensitive record identifier, it is -0.0010 and not significant.

Figure 6 shows the two failure modes at the level of a single case, with the predictions the reader actually produced.

The two-hop column is the most informative single result in this paper. Under the delete variant the reference answer is present in the context in 96.3% of cases, so GIC is essentially saturated, and F1 is 0.0000. The information is in the prompt; the reader cannot use it, because the identifier saying which record it belongs to has been removed. String-presence metrics do not detect this failure, which is why we report task quality and leakage as separate endpoints rather than as one utility score.

8.5 Question-side aliasing

Aliasing the context without rewriting the question produces a prompt whose halves use different names for the same entity. On the 112-case set we measured both settings with the pipeline otherwise unchanged: `alias` moves from 0.4933 to 0.6719 ($+0.1786$) and `alias_budget_link` from 0.5048 to 0.7450 ($+0.2402$). The step costs nothing in privacy, since both halves stay free of the raw value, and it is the difference between a prompt the reader can use and one it cannot. We therefore treat the pair (q', P) rather than P alone as the object of study, and verify the mapping before generation.

8.6 The direction of the refill traversal

Section 4.5 noted that the traversal direction for refill units is an empirical choice. We ran both variants end to end.

Benchmark	increasing	decreasing	Δ	95% CI
extension (2,728)	0.4937	0.4832	-0.0106	$[-0.028, +0.007]$
main set (112)	0.6366	0.5844	-0.0522	$[-0.124, +0.019]$
topical (300)	0.3516	0.3172	-0.0344	$[-0.090, +0.020]$
single-hop (300)	0.9607	0.9719	$+0.0111$	$[-0.008, +0.032]$
two-hop (300)	0.6733	0.0167	-0.6567	$[-0.712, -0.599]$

For four of the five rows the interval includes zero. Only the two-hop benchmark separates the two variants, and it separates them completely. There the answer sits six sentences from the anchor, outside the fallback window, so whether the answer is present at all is decided by the refill order; keeping repetitive units happens to retrieve it. We report this as what it is: an empirical regularity of this data, consistent with the anti-correlation measured at the start of this section, but not a principled guarantee. The principled remedy is to widen the relevant region until it covers the answer, which Section 9 takes up.

8.7 Where the rule applies

Running the same pipeline on the three LongBench-derived variants locates the boundary.

Variant	`full`	`alias`	DARK
topical (300)	0.4181	0.4074	0.3516
single-hop (300)	0.9889	0.9197	0.9607
two-hop (300)	0.0967	0.7078	0.6733

On the topical variant the questions name no entity, so there is nothing to route on; the rule degenerates to global aliasing plus redundancy ordering, and it loses 0.0665 against the uncompressed prompt. We report that variant as out of scope, not as a success.

The two-hop row carries a second observation we did not anticipate. Without sanitization the raw prompt scores 0.0967; with global aliasing it scores 0.7078. The account identifiers in this variant are long hexadecimal strings differing in a few characters, and replacing them with short

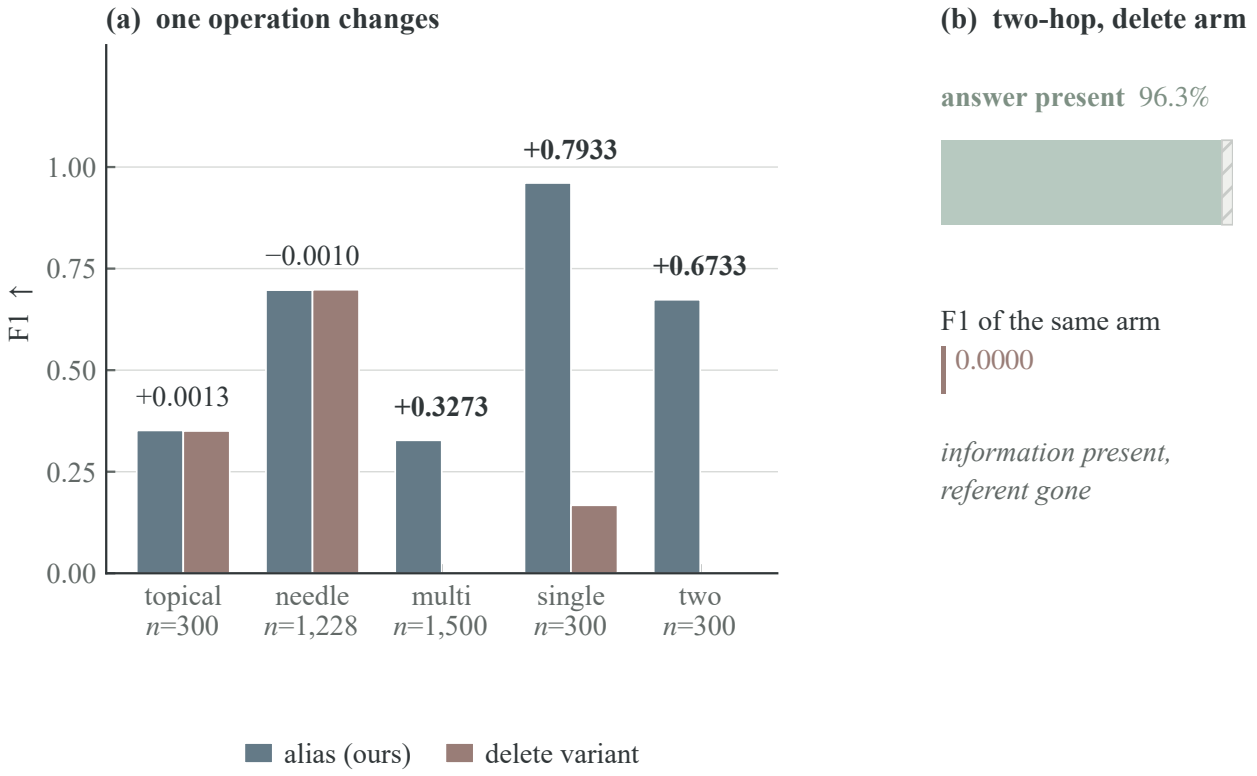


Figure 5: Replacing the alias operation by deletion, with the context construction otherwise unchanged, on five settings. (a) The difference concentrates exactly where the anchor is a protected value and vanishes where it is not; labels give the paired difference of the two bars and a star marks an interval that excludes zero. (b) Distribution of the paired per-case difference for the three settings where the operation matters. The mass sits at both ends because a case is either answered or not, and the mean gain is largest on the single-hop variant. (c) In the two-hop variant the answer is still in the prompt after 96.3% of the deletions, and F1 is zero.

distinct tokens makes the task *easier* for the reader. Privacy and utility move in the same direction here, which is a useful counterpoint to the usual framing but also a reminder that the effect belongs to the identifier surface form, not to the routing rule.

9 Discussion

The scope the evidence supports. The claim we support is bounded. For excerpts of IoT-style operational records, queried by a question that names an entity whose value must not reach the model, routing each unit on (sensitivity, relevance) and aliasing the sensitive-and-relevant cell preserves the task at half the tokens and at zero measured leakage. The three variants in Section 8 delimit it: the rule is unnecessary when the question is topical, redundant when anchor and answer coincide in a single record, and dependent on the refill order when the answer lies outside the fallback window. We do not claim that it improves ordinary compression on ordinary prompts, and the topical variant shows that it does not.

What the privacy claim covers. Leakage is measured as recoverability of the protected values from the exact prompt text, using the pattern set of the published protocol (full value plus fragments). That is the relevant quantity for a pipeline whose adversary is the model provider and

(a) two-hop: the answer sits behind a chain

Q: which device does account `ACCT-5f31` operate?

```
164  account ACCT-5f31 belongs to user U-5a675c
...
170  user U-5a675c operates device sensor_422708_15
```

after aliasing, both links keep a name:

```
account [ACCT-A] belongs to user [USER-B]
user [USER-B] operates device sensor_422708_15
```

(b) multi-entity: the answer looks like a distractor

Q: which persona id is recorded for account `ACCT-447f...`?

```
07  {"device": "dehumidifiers",
     "id": "multi_0166_00", ...}
08  Private contact record:
09  Contact email: contact.e67e...
11   Account ID: ACCT-447fa54ac66d
```

the target account is the eleventh of sixteen,
and its record is shaped like every other record.

	full	alias	delete	DARK
(a) two-hop, `lbc-hotpotqa-0000`	sensor_422708_15	sensor_422708_15	UNKNOWN	sensor_422708_15
(b) multi-entity, `multi-0032`	UNKNOWN	multi_0166_00	UNKNOWN	multi_0166_00

Figure 6: Two cases where the routing changes the answer, with the predictions the reader actually produced. In (a) the question names an account that reaches the answer only through a user identifier six records later; deleting either link leaves the answer present but unreachable, and the model returns UNKNOWN. In (b) the answer sits in one of sixteen near-identical account records, so the uncompressed prompt *also* fails—here the distractor structure, not the budget, is what defeats the reader—while both aliasing arms answer correctly. Shaded spans mark the names that keep a chain addressable.

whose prompts may be retained. It is not a formal guarantee. We derive no differential-privacy bound [3] and no k-anonymity guarantee [14]; the alias map is assumed private, and an adversary who obtains it recovers the values. We fix the alias assignment from a keyed permutation precisely so that an adversary holding only the value set cannot invert the map by rank, which is a real attack on unkeyed assignment, but that mitigation is narrower than a guarantee. We also do not evaluate inference attacks—membership, attribute, or the reconstruction of a value from its alias's co-occurrence pattern—and none of them is addressed by the present evidence.

The refill order is a regularity. The measured benefit of keeping repetitive units follows from an anti-correlation between redundancy and answer-bearing in this data. It is not derived from the definition of the score, and a corpus whose answers are unique rather than repeated would not exhibit it. The principled fix is to make the relevant region cover the answer instead of relying on the ordering. In the two-hop variant the answer lies at a fixed distance of six sentences from the anchor, so a window of that width would protect it explicitly and remove the dependence. We have not run that configuration, and the present results should be read as depending on the fallback width.

Cost. Prompt construction is lexical and model-free: one pass over the units to label them, one pass to score redundancy (quadratic in the number of units in the worst case), and one pass to fit the budget. It adds no model call. On the longest excerpt in the extension set (19,838 tokens uncompressed) construction completes in under a second on CPU. We did not measure end-to-end wall-clock or memory of the reader, so the efficiency we claim concerns prompt length rather than latency.

Threats to validity. The protected values are synthetic canaries injected into public corpora, because those corpora cannot carry real personal data. Canary leakage is measurable exactly, but the measured rate is a property of this construction. Both benchmark sizes are generated by our own scripts from shared templates, so the questions are entity-resolving by construction and the two-hop distance is fixed rather than sampled. The readers are 7B-parameter instruction-tuned models, and we have no evidence about larger or API-served models. Finally, the two benchmark

sizes reuse case identifiers for different content, so a cross-set join by identifier would silently mix records; we guard against this by verifying the uncompressed arm against a per-set reference value, but the hazard is worth stating for anyone reusing the artifacts.

10 Conclusion

We studied what happens to a privacy-preserving inference pipeline when the entity a question asks about is also the entity that must be protected. Two standard remedies fail there, for opposite reasons. Redaction removes the referent the reader needs. Redundancy ordering removes the record that carries the answer, because on this data the answer-bearing record is the most repetitive one. We presented DARK, which resolves the conflict by deciding the two axes per unit and by giving the sensitive-and-relevant cell an operation that neither remedy provides: the unit stays, its values are aliased, and the same map is applied to the question. Relevance is read off the question—the record block that mentions the queried entity—so the pipeline needs neither a reference answer nor training.

Across a 112-case benchmark over six public datasets, DARK uses 48.6% of the tokens at zero prompt-level leakage and is statistically indistinguishable from aliasing the full context without compression (-0.0109). On a $2{,}728$-case extension it matches the strongest query-aware compressor ($+0.0107$, not significant) at 0% leakage against 98.1%, and exceeds the nearest leak-free baseline by 0.3226 F1 points. A single controlled substitution isolates the decision behind these numbers: replacing the alias operation with deletion costs 0.1795 F1 points overall, 0.3273 on multi-entity cases, and nothing where no anchor exists. The last figure matters as much as the others, because it indicates that the rule acts on the structure it was designed for and is inert elsewhere.

Two directions follow from the limits stated in Section 9. The fallback window is the weakest part of the mechanism: in the two-hop variant the answer lies outside it and is recovered only through the refill order, so widening the relevant region to cover the answer is the first configuration to test. The second concerns the privacy claim itself. A pipeline of this shape would be considerably stronger with a guarantee that survives an adversary holding the alias map, and establishing one is a different piece of work from the one reported here.

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Table 5: Per-source F1 on the extension set (2,728 cases, $r = 0.50$).

Source	n	full	DARK	ABL
pec_home	426	0.7816	0.6756	0.2859
privacylens	600	0.5340	0.4671	0.3381
simuhome	312	0.8686	0.6186	0.3910
smartbench_tdcs	490	0.6449	0.5714	0.6367
toolmem	300	0.5154	0.4933	0.3121
toolmem_raw	600	0.4401	0.2631	0.1624
overall	2728	0.6081	0.4937	0.3482

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A Appendix

A.1 Per-source breakdown

Extension set (2,728 cases). F1 by source, budget $r = 0.50$, all arms of ours constructed without the answer key. The routing rule is strongest on pec_home and smartbench_tdcs and weakest on toolmem_raw, where the uncompressed arm is also weakest.

Main set (112 cases). Same setting on the smaller set.

Table 6: Per-source F1 on the main set (112 cases, $r = 0.50$).

Source	n	full	DARK	alias
pec_home	20	0.8750	0.8250	0.8250
privacylens	20	0.6567	0.6400	0.4900
simuhome	20	0.8500	0.7500	0.7500
smartbench_tdcs	20	0.7000	0.7000	0.6500
toolmem	12	0.4901	0.4167	0.4354
toolmem_raw	20	0.6500	0.4000	0.6500
overall	112	0.7189	0.6366	0.6475

A.2 Anchor types covered

The anchor key is resolved differently depending on what the question names. All six forms occurring in the extension set are handled, and the anchor string is present in the context in every case ($2{,}728/2{,}728$).

Table 7: Anchor key forms covered by the routing rule. All six occur in the extension set and the anchor string is present in the context in every case ($2{,}728/2{,}728$).

anchor_kind	Example question form	n
none (entity field)	*for account* ACCT-...	1500
id	*for record id* pec_home_...	316
name	names an entity directly	300
session	names a session	300
entry	*for entry 3*	281
native	native identifier	31

A.3 Leakage counting

A case counts as leaking when any protected value, or a fragment of it, is recoverable from the concatenation of the rewritten question and the compressed context. Fragments are the pattern set of the published protocol. The count is per case, not per value, so the reported rate is the fraction of cases with at least one exposure; the per-value count is retained in the artifacts.

A.4 Reproduction notes

Every table in this paper is computed from per-case records that ship with the artifacts, one line per (case, arm). The uncompressed arm in each set is used as an identity check: a result set whose uncompressed arm does not reproduce the reference value is not part of this paper. Budget arms assert non-overshoot per case; a run that violates it is discarded rather than truncated. All arms of ours are keyed alias runs; an unkeyed assignment changes both the alias strings and the resulting score, so results are only comparable within one setting.